ML for Text Mining

Section 4

	The	old	man	helped	the	small	cat
You have Done POS Tagging:	DET	JJ	NN	VBD	DET	JJ	NN
You have Done Dependeny Parsing:			nsubj	VERB			dobj

Now, You can do Ontology-Based semantic Information Extraction (OBIE)

Ontology

Formal specification of:

- **Entities**
- **Properties and Relationships between them**

in specific domain

What is domain?

Medical, Finance, Social network, Genes, or any document itself can be a domain

What are entities?

In medical: Drug, Disease, Patient, …

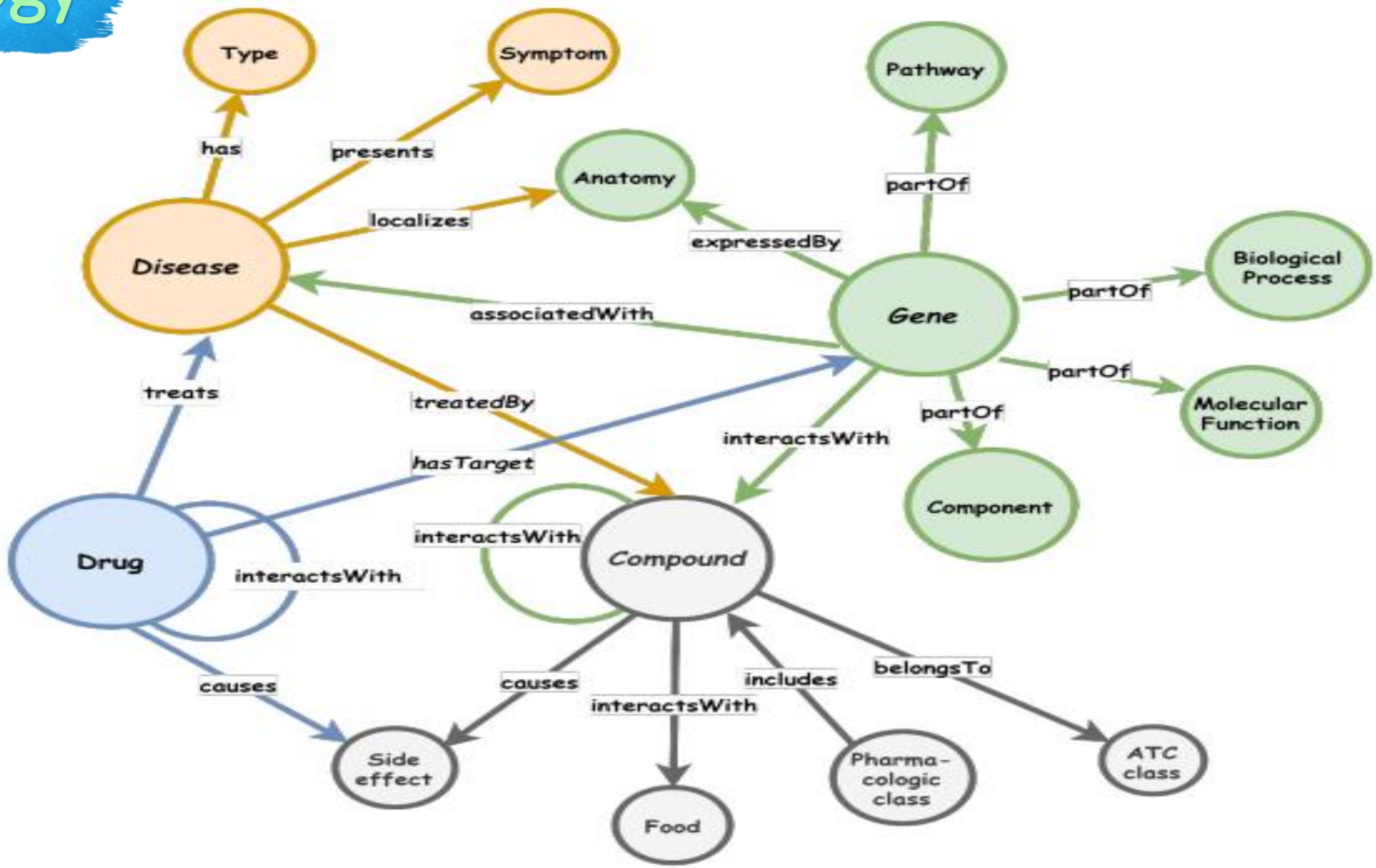
What are Properties or relations?

In medical: Has_tradename, Type, Has_Stage, gradient_of, cure...

In social network: gender, email, has_occupation, friend of ..

Formal, so there is no specific values for entities or properties!

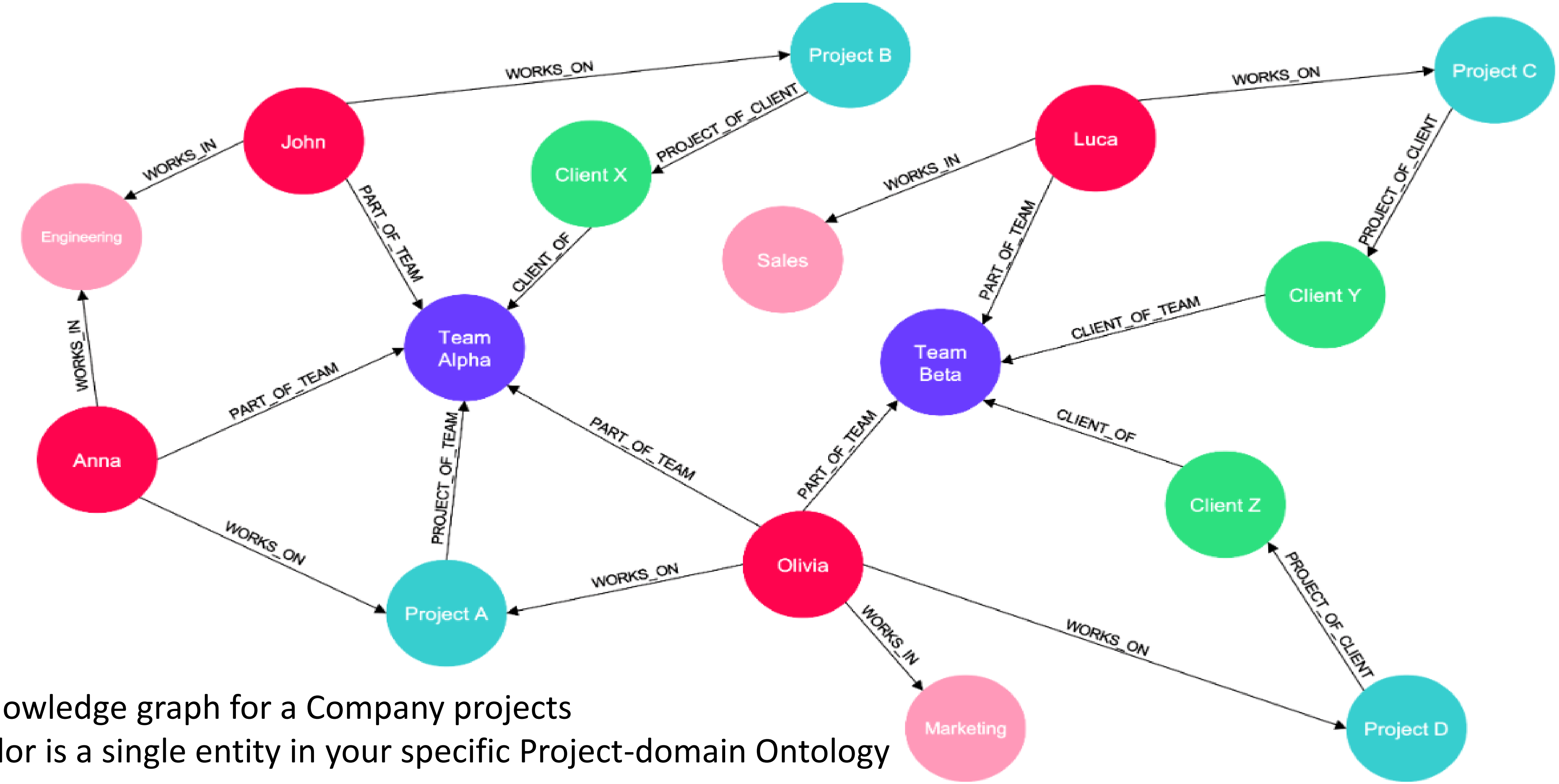
Ontology



This is your **Ontology** (Structure of entities and relations) BAM!

Examples

Apply it to your interesting **Documents** and you will get a **Knowledge Graph**.



Example for: knowledge graph for a Company projects
Each unique color is a single entity in your specific Project-domain Ontology

ALERT!

Don’t think like “Sure! This is a dead useless Methodology”

GraphRAG (2024 by Microsoft) uses Knowledge Graphs in its pipeline

ALERT 2

An example of a semantic knowledge base is________

Examples

Gene Ontology

(Genetics)

DBpedia

(Wikipedia blogs)

Schema.org

(General)

Focused on understanding of domain entities and their relationships

Wordnet

Relation between Language terms

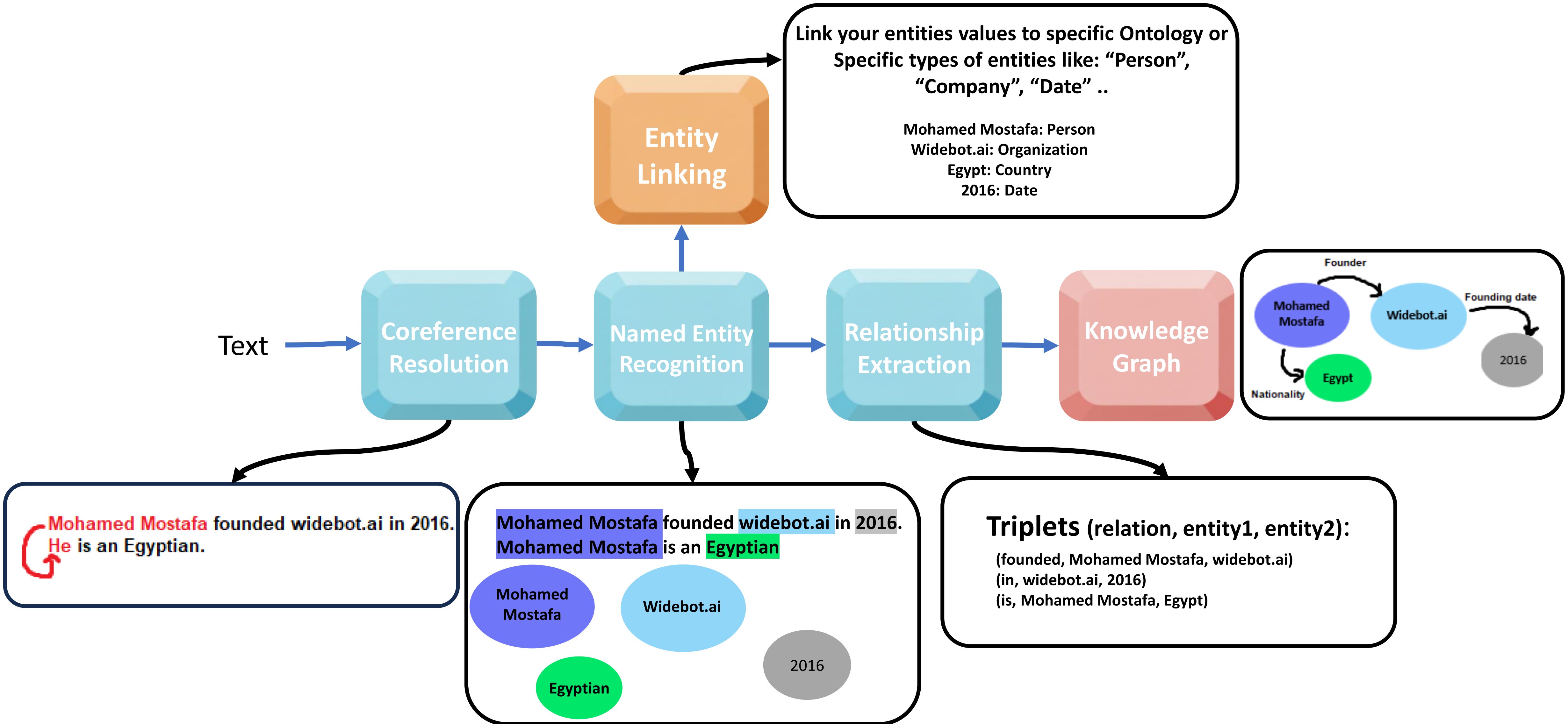
_ is a _	_ part of _	_ synonym of _
animal is a organism	head part of animal	Animal syn of being

Focused on Conceptual interpretation of words in linguistic form

KG Build

This days method is to use LLM to create the knowledge graph, we will explain it at the end, but for now let’s take a look of how this can be done using NLP and rule-based processing steps.

How to build Knowledge Graph in general



KG Build

In Current Era, We actually use LLMs to Build Knowledge Graphs (KGs)

By passing to them our text or documents along with 2 types of input

Few Shot Prompting

```
"""You are a top-tier algorithm designed for extracting
information in structured formats to build a knowledge
graph. Your task is to identify the entities and relations
requested with the user prompt from a given text. You must
generate the output in a JSON format containing a list with
JSON objects. Each object should have the keys: "head",
"head_type", "relation", "tail", and "tail_type". Here is
one example:
---
Give the text: "Adam is a software engineer in Microsoft
since 2009"
You can extract a relationship in the following format:
{
  "head": "Adam",
  "head_type": "Person",
  "relation": "WORKS_FOR",
  "tail": "Microsoft",
  "tail_type": "Company"
}
```

One shot

Multiple runs may result in Different outputs

Structured Output

```
class Node(BaseNode):
    id: str
    label: str
    properties: Optional[List[Property]]

class Relationship(BaseRelationship):
    source: Node
    target: Node
    type: str
    properties: Optional[List[Property]]

class KnowledgeGraph(BaseModel):
    """Generate a knowledge graph with entities and relationships."""
    nodes: List[Node] = Field(
        ..., description="List of nodes in the knowledge graph"
    )
    rels: List[Relationship] = Field(
        ..., description="List of relationships in the knowledge graph"
    )
```

Entity definiton

Relationship definition

The needed structure form of the output

More Consistency in output

Resources: 1 & 2

KG Build

By asking one of my friends that works as AI Engineer in a company, he stated that they use LLMs to build Knowledge graphs in the same way.

Thank You